Last week:

- Measuring connectivity:
  - Cut-vertices (delete a vertex, get more components)
  - Blocks (decompose any connected graph as a union of non-separable subgraphs which can only intersect at cut-vertices)
  - Vertex-cuts (delete a set of vertices, get more components)

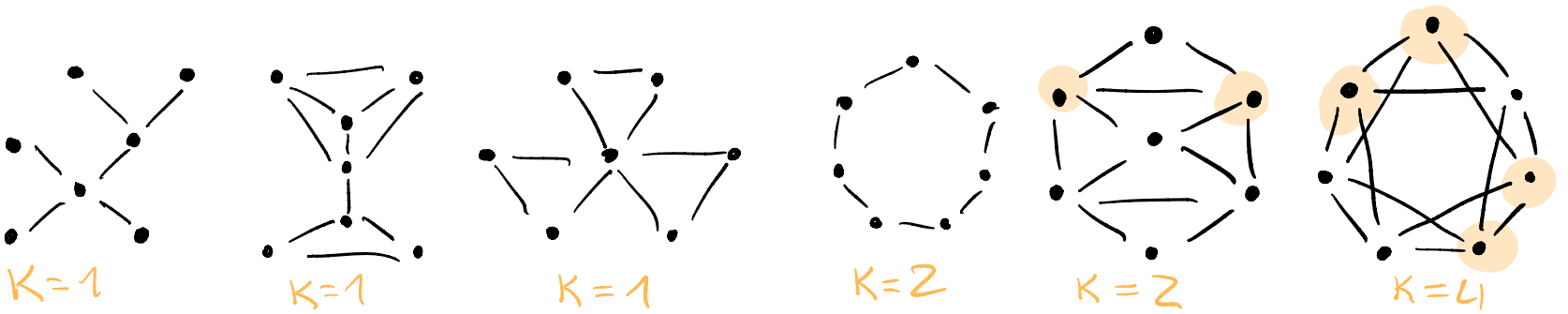
This week:

- Measuring connectivity:
  - Edge-cuts
  - Vertex- and edge-connectivity
- Menger's theorem.

Recall: Let G be a graph

- A **vertex-cut** is a subset $U \subset V(G)$ such that $G - U$ is disconnected.
- The **vertex-connectivity** $\kappa(G)$ is the minimum cardinality of a vertex-cut.
- For $G = K_n$, we set $\kappa(G) = n - 1$ (because it contains no vertex-cuts).
- If k is a non-negative integer, we say that G is k-connected if $\kappa(G) \geqslant k$.

Example. *Compute* $\kappa(G)$ *for the following graphs.*



$\lambda(K_n) = n-1$

Definition. *Let* G *be a graph*

- *An* ***edge-cut*** *is a subset* $F \subset E(G)$ *such that* $G - F$ *is disconnected.*
- *The* ***edge-connectivity*** $\lambda(G)$ *is the minimum cardinality of an edge-cut. For* $G = K_1$*, we set* $\lambda(G) = 0$*.*
- *If* k *is a non-negative integer, we say that* G *is* k*-edge-connected if* $\lambda(G) \geqslant k$*.*

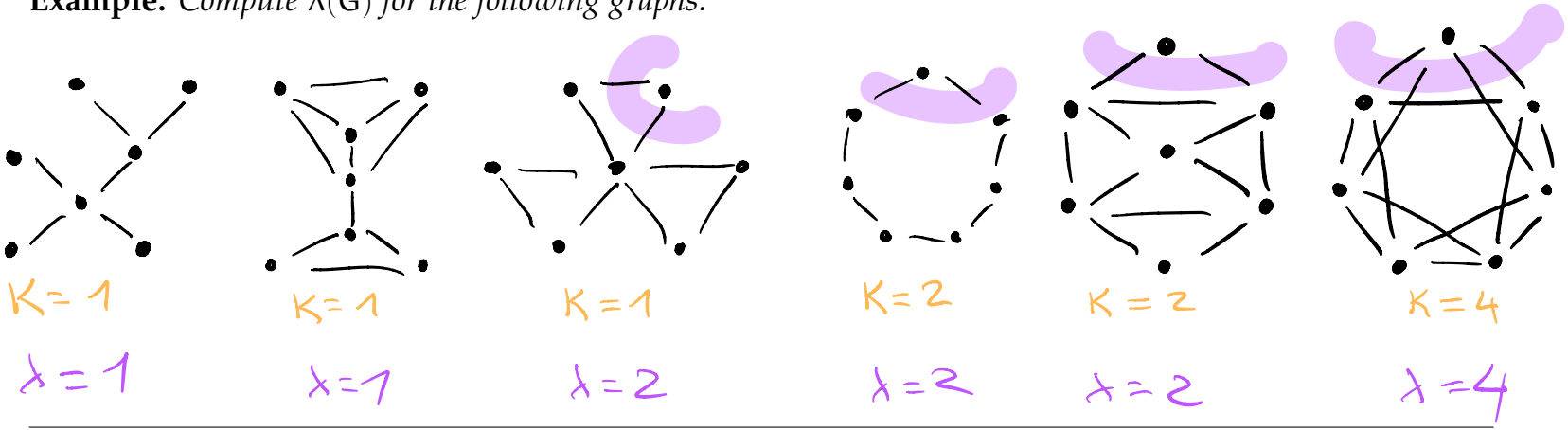
Note: : Assume G has n vertices.

G is connected and

(1) $0 \leqslant \lambda(G) \leqslant n-1$ (2) $\lambda(G) = 0$ iff G disconnected (3) $\lambda(G) = 1$ iff G has a bridge.

or $G = K_1$

Example. *Compute* $\lambda(G)$ *for the following graphs.*



Note: Assume G connected

Let F minimum edge cut. Thus $G-F$ disconnected.

Then 1) $G-F$ has exactly two components G_1 and G_2

2) Any edge in F goes between G_1 and G_2

Proposition (5.10). *We have* $\lambda(K_n) = n - 1$ *for any positive integer* n,

Proof: For $n = 1$, by definition, $\lambda(K_1) = 0$. Assume $n \geq 2$.

Note: $\lambda(K_n) \leq n-1$: Fix a vertex v, then all edges incident to v form an edge-cut with $n-1$ elements.

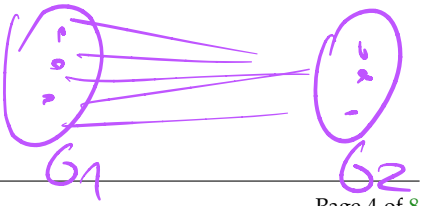
Take a minimum edge cut F. Thus $\lambda(K_n) = |F|$. WTS: $|F| \geq n-1$

1) $K_n - F$ has two components G_1, G_2

Say $k = |V(G_1)|$, and $n-k = |V(G_2)|$, $k \geq 1$, $n-k \geq 1$.

2) F uses all edges between G_1 and G_2

Thus $|F| = k(n-k)$



WTS: $k(n-k) \geq n-1$

$k \geq 1, \quad n-k \geq 1 \Rightarrow k-1 \geq 0, \; n-k-1 \geq 0$

$\Rightarrow (k-1)(n-k-1) \geq 0$

$0 \leq k(n-k) - k - n + k + 1 = k(n-k) - n + 1$

$\Rightarrow n-1 \leq k(n-k)$

So $|F| \geq n-1 \Rightarrow |F| = n-1 \Rightarrow \lambda(K_n) = n-1.$

Theorem (5.11). *For every graph* G *we have* $\kappa(G) \leqslant \lambda(G) \leqslant \delta(G)$.

Proof: If G is disconnected, Then $\kappa(G) = 0 = \lambda(G) \leq \delta(G)$

If $G = K_n$, $\kappa(G) = n-1 = \lambda(G) = \delta(G)$

Thus we can assume G is connected not complete on $n \geq 3$ vertices. Thus $\delta(G) \leq n-2$

STEP 1: $\lambda(G) \leq \delta(G)$

Let v in $V(G)$ with $\deg_G v = \delta(G)$. Then $E_v = \{e \in E(G) : v \in e\}$

Then E_v is an edge cut with $|E_v| = \delta(G)$ elements

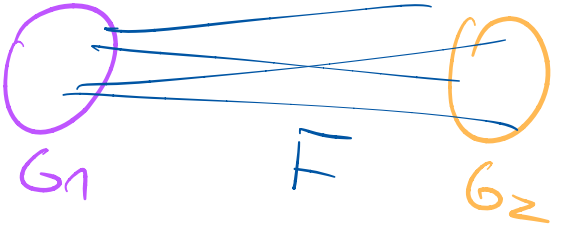
STEP 2: $\kappa(G) \leq \lambda(G)$.

Let F minimum edge cut. $|F| = \lambda(G) \leq n-2$ Step 1

F minimum: 1) $G-F$ has exactly two components G_1, G_2

Say $K = |V(G_1)|$, $n-K = |V(G_2)|$

2) Every edge in F goes between G_1 and G_2



Case 2.1: Every vertex in G_1 is adjacent to every vertex in G_2

Then $|F| = K(n-K)$ (previous argument)

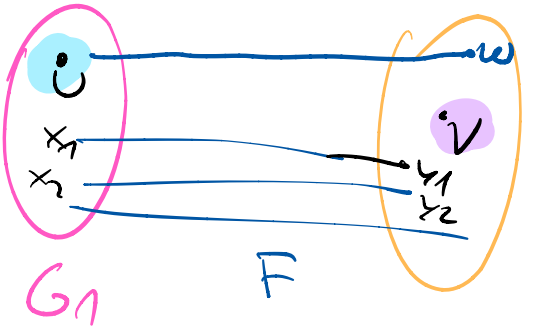
Then $\lambda(G) = |F| = K(n-K) \geq n-1$

Contradicts $\lambda(G) \leq n-2$.

Case 2.2: $\exists\, u \in V(G_1)$, $v \in V(G_2)$ such that $uv \notin E(G)$

Next: extract from F a vertex cut U

Let $e \in F$ → If e uses $v \Rightarrow e = vw$ for some $w \in V(G_2)$, put w in U



If $u \notin e \Rightarrow e = xy$, $x \in V(G_1)$, $y \in V(G_1)$ and $x \neq u$

Put x in U

Note that:

1) By construction $|U| \leq |F|$

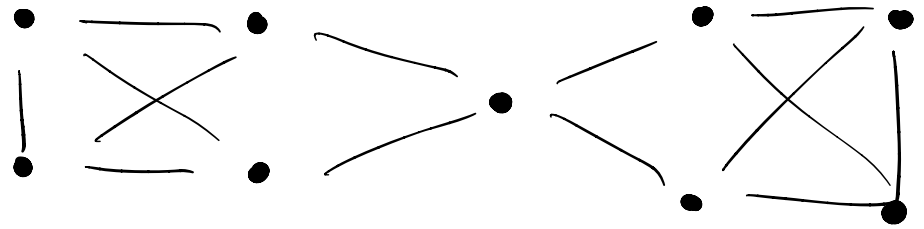
2) $u, v \notin U$

3) There is no u-v path in $G-U$ (we deleted all edges between G_1 and G_2). Hence $G-U$ disconnected

Then U is a vertex cut

$$\kappa(G) \leq |U| \leq |F| = \lambda(G)$$ □

Example. *The bounds in Withney's theorem are sharp:*



$\kappa(G) = 1$
$\lambda(G) = 2$
$\delta(G) = 3$

Example. *If* G *is* r*-regular, then* $\kappa(G) \leqslant r$.

Corollary (5.13). *If* G *has* n *vertices and* m *edges, then* $\kappa(G) \leqslant \left\lfloor \frac{2m}{n} \right\rfloor$.

Proof: $\delta(G) \leqslant \left\lfloor \frac{2m}{n} \right\rfloor$

Otherwise $\deg v > \frac{2m}{n}$ for all v

Then $2m = \sum_{v \in V(G)} \deg v > \sum_{v \in V(G)} \frac{2m}{n} = 2m$ Contradiction.

$\Rightarrow \delta(G) \leqslant \frac{2m}{n} \Rightarrow \delta(G) \leqslant \left\lfloor \frac{2m}{n} \right\rfloor$

- A **cubic** graph is a 3-regular graph

Theorem (5.12). *If* G *is a cubic graph, then* $\kappa(G) = \lambda(G)$.

Exercise. *Compute* $\kappa(G)$ *and* $\lambda(G)$ *for the Petersen graph.*

Idea: $\kappa = 0 \iff$ disconnected $\iff \lambda = 0$

$\kappa = 3 \iff$ complete $K_4 \iff \lambda = 3$

$\kappa = 1$, vertex cut $U = \{u\}$

$G - U$

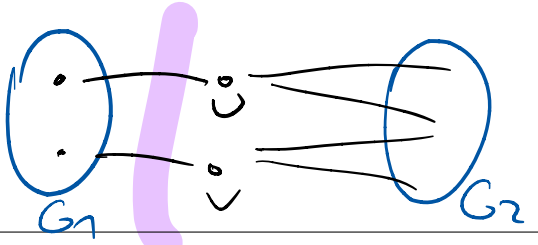


$\exists$ component of $G - U$ with a unique neighbor u' of $u \Rightarrow uu'$ bridge

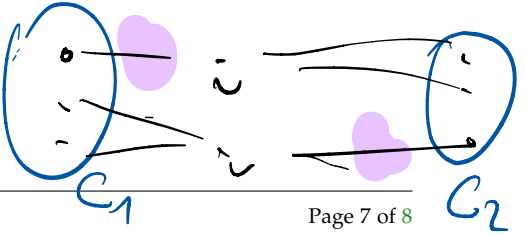
$\kappa = 2$

$U = \{u, v\}$

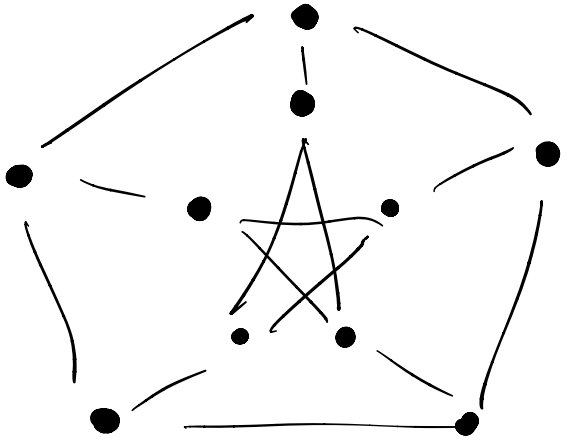
case 1



Case 2:



Example



3-regular $\Rightarrow$ $\kappa = \lambda$

$\downarrow$

not bad!

$\lambda \neq 0$ (connected)

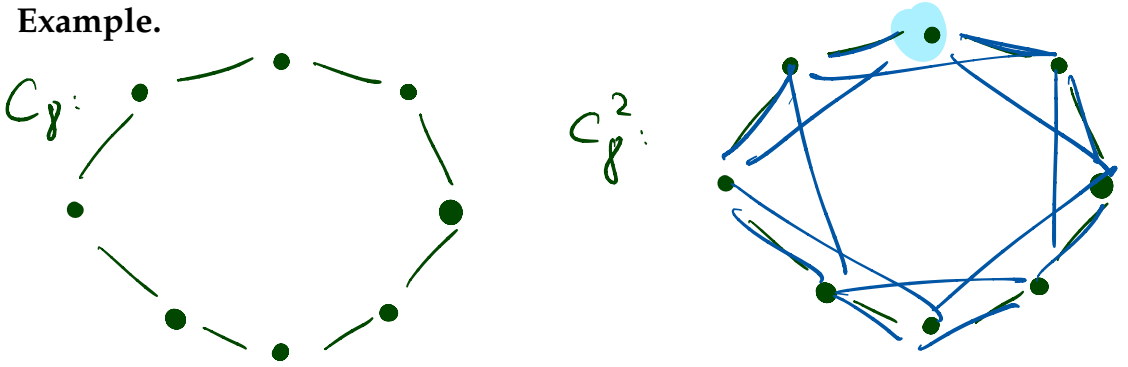
$\lambda \neq 1$ (no bridges)

$\lambda = 2$?

Compute K

Definition. *Let G be a graph and k a positive integer. The **k-th power** of G is the graph G^k which has the same vertex-set as G and an edge $uv \in E(G^k)$ iff $d_G(u,v) \leqslant k$.*

Example.



Example (Harary graphs). *For r, n not both odd and $1 \leqslant r \leqslant n-1$, we defined $H_{r,n}$, an r-regular graph on n-vertices $\{v_1, \ldots, v_n\}$.*

- *If $r = 2k$ is even, then $H_{r,n} = C_n^k$.*
- *If $r = 2k+1$ is odd and $n = 2\ell$ is even, then $H_{r,n} = C_n^k \cup \{v_i v_{i+\ell} : 1 \leqslant i \leqslant \ell\}$.*

We have $\kappa(H_{r,n}) \leqslant \delta(H_{r,n}) = r$. Can we show $\kappa(H_{r,n}) = r$?

Exercise. *G^k is a complete graph iff $diam(G) \leqslant k$.*

Exercise. *If G is a connected graph on at least $k+1$ vertices, then G^k is k-connected.*